STRING

Strings in python are surrounded by either single quotation marks, or double quotation marks.

index start with 0.

In [6]:
```
# get user input
name=input("enter name:")
print(name)
```

```
Reeta
```

In [2]:
```
print('hi')
```

```
hi
```

In [10]:
```
k="Hello Every one"
print(k)
```

```
Hello Every one
```

Multiline string

In [2]:
```
k=""" Late at night, Emma found an old,
    dusty key hidden under a loose floorboard.
    Curiosity led her to the attic,
    where she discovered a locked chest that seemed to hum with energy.
    As she turned the key, the chest creaked open,
    revealing a portal to a world she never imagined."""

print(k)
```

```
 Late at night, Emma found an old,
    dusty key hidden under a loose floorboard.
    Curiosity led her to the attic,
    where she discovered a locked chest that seemed to hum with energy.
    As she turned the key, the chest creaked open,
    revealing a portal to a world she never imagined.
```

In [39]:
```
print("It's alright")
print("He is called 'Johnny'")
print('He is called "Johnny"')
```

```
It's alright
He is called 'Johnny'
He is called "Johnny"
```

1.string length

In [12]:
```
k="Hello Every one"
print(len(k))
```

```
15
```

In [8]:
```
name = "Reeta"
print(len(name))
```

```
5
```

2.string indexing

In [9]:
```
b = "Hello,  World!"
print(b[-4])
```

```
r
```

In [18]:
```
b = "Hello, World!"
print(b[4])
```

```
o
```

3.string slicing

In [24]:
```
b = "Welcome everyone"
print(b[-7:-1])
```

```
veryon
```

In [26]:

```
b="python"
b[1:]
```

Out[26]:

```
'ython'
```

In [28]:

```
b = "Hello, World!"
print(b[2:])
```

```
llo, World!
```

In [30]:

```
b = "Hello, World!"
print(b[:6])
```

```
Hello,
```

In [32]:

```
b = "Hello, World!"
print(b[7:12])
print(b[-6:-1])
```

```
World
World
```

In [34]:

```
b = "welcome, World!"
print(b[-5:-2])
```

```
orl
```

In [36]:

```
b = "Hello, World!"
print(b[-5:])
```

```
orld!
```

4.string modify

upper

lower

capitalize

strip

split

replace

In [18]:

```
a = "Hello, World!"
print(a.upper())
```

```
HELLO, WORLD!
```

In [41]:

```
a = "heLlo, World!"
print(a.capitalize())
```

```
Hello, world!
```

In [43]:

```
a = "Hello, World!"
print(a.lower())
```

```
hello, world!
```

In [45]:

```
a = "    Hello, World!    "
print(a)
print(a.strip())
```

```
    Hello, World!
Hello, World!
```

In [47]:

```
a = "    Hello, World!    "
b=a.strip()
print(len(a))
print(len(b))
print(b)
```

```
21
13
Hello, World!
```

In [53]:

```
a= "Hello World"
print(a.split("W"))
```

['Hello ', 'orld']
In [55]:
```
a="hi  hello"
print(a.split())
```
['hi', 'hello']
In [59]:
```
a = "Hello, World!"
print(a[0])
b=a.split()
print(b)
print(b[0])
```
H
['Hello,', 'World!']
Hello,
In [63]:
```
#replace
a = "Hello, World!"
print(a.replace("H", "J"))
```
Jello, World!
In [68]:
```
a="I love java"
a.replace("java","Python")
```
Out[68]:
'I love Python'

5.formate

In [61]:
```
a=input("enter your name:")
b=input("enter your place:")
k=f"i am {a} from {b}"
print(k)
```
i am Reeta from Salem

6.String Concatenation

In [72]:
```
a="Reeta"
b="Elayaraja"
full_name = a+b
print(full_name)
```
ReetaElayaraja
In [74]:
```
a="Reeta"
b="Elayaraja"
print(a+" "+b)
```
Reeta Elayaraja

7.Escape Characters

1.Single Quote

In [82]:
```
a="It\'s me"
print(a)
```
It's me

2.Backslash

In [89]:
```
a="Bigg boss is 24\\7 live"
print(a)
```
Bigg boss is 24\7 live

3.New Line

In [92]:
```
a="Reeta\nElayaraja"
print(a)
```
Reeta
Elayaraja

4.Carriage Return

In [95]:
```
a="Reeta\rElayaraja"
print(a)
```
Elayaraja

5.Tab

In [98]:
```
a="Reeta\tElayaraja"
print(a)
```
Reeta Elayaraja

6.Backspace

In [105]:
```
a="Reeta  \bElayaraja"
print(a)
```
Reeta Elayaraja

8.String Methods

1.capitalize

In [111]:
```
a="the art of begin alone"
b=a.capitalize()
print(b)
```
The art of begin alone

2.casefold

In [124]:
```
a="REETA"
b=a.casefold()
print(b)
```
reeta

3.Upper

In [127]:
```
a="reeta"
b=a.upper()
print(b)
```
REETA

4.Lower

In [122]:
```
a="REETA"
b=a.lower()
print(b)
```
reeta

5.center

In [141]:
```
a="Reeta"
b=a.center(10)
print(b)
```
  Reeta

6.count

In [144]:
```
a="The art of begin alone"
b=a.count("a")
print(b)
```
2

7.encode

In [147]:
```
a="Reeta"
b=a.endswith("a")
print(b)
```
True

8.startswith

In [152]:
```
a="Reeta"
b=a.startswith("R")
print(b)
```
True
In [154]:
```
a="Reeta"
b=a.startswith("r")
print(b)
```
False

9.expandtab

In [157]:
```
a="R\te\te\tt\ta"
b=a.expandtabs(1)
print(b)
```
R e e t a

10.find

In [160]:
```
a="The art of begin alone"
b=a.find("art")
print(b)
```
4
In [162]:
```
a="The art of begin alone"
b=a.find("o")
print(b)
```
8

11.index

In [165]:
```
a= "Hello, welcome to my world."
b=a.index("world")
print(b)
```
21

12.isalnum

In [168]:
```
a="123abc"
b=a.isalnum()
print(b)
```
True

13.isalpha

In [173]:
```
a="abc"
b=a.isalpha()
print(b)
```
True

14.isdecimal

In [178]:
```
a="2343"
b=a.isdecimal()
print(b)
```
True

15.isdigit

In [187]:

```
a="002343"
b=a.isdigit()
print(b)
```

```
True
```

16.isidentifier

```
In [194]:
a="demo"
b=a.isidentifier()
print(b)
```

```
True
```

```
In [196]:
a="2demo"
b=a.isdecimal()
print(b)
```

```
False
```

```
In [198]:
a="my demo"
b=a.isdecimal()
print(b)
```

```
False
```

17.islower

```
In [201]:
a="reeta"
b=a.islower()
print(b)
```

```
True
```

18.isupper

```
In [206]:
a="REETA"
b=a.isupper()
print(b)
```

```
True
```

19.isnumeric

```
In [209]:
a="2455"
b=a.isnumeric()
print(b)
```

```
True
```

20.isprintable

```
In [214]:
a= "Hello! Are you?"
x = a.isprintable()
print(x)
```

```
True
```

21.isspace

```
In [225]:
a=" "
b=a.isspace()
print(b)
```

```
True
```

22.title

```
In [230]:
a="the art"
b=a.title()
print(b)
```

```
The Art
```

23.istitle

```
In [237]:
```

```
a="The Art"
b=a.istitle()
print(b)
```
True

24.join

In [242]:
```
a="Reeta","Mano"
b="".join(a)
print(b)
```
ReetaMano

25.ljust

In [245]:
```
a="apple"
b=a.ljust(4)
print(b,"is my fav friut")
```
apple is my fav friut

26.lstrip

In [248]:
```
a="   Reeta"
b=a.lstrip()
print(b)
```
Reeta

27.replace

In [251]:
```
a="Reeta"
b=a.replace("Reeta","Mano")
print(b)
```
Mano

28.translate

In [2]:
```
a = "abc aab"
b = str.maketrans("a", "e")
print(a.translate(b))
```
ebc eeb

29.formate_map

In [5]:
```
a={"name":"reeta","age":20}
b="My name is {name} and I am {age} old."
print(b.format_map(a))
```
My name is reeta and I am 20 old.

30.zfill

In [18]:
```
a="10"
b="Reeta"
print(a.zfill(10))
print(b.zfill(10))
```
0000000010
00000Reeta

31.swapcase

In [21]:
```
a="I Love Python Program"
print(a.swapcase())
```
i lOVE pYTHON pROGRAM

32.encode

In [24]:
```
a="Hello"
print(a.encode())
```
b'Hello'

33.removeprefic

In [28]:
```
a="reeta.mano"
print(a.removeprefix("reeta."))
```
mano

34.removesuffic

In [37]:
```
a="reeta.mano"
print(a.removesuffix(".mano"))
```
reeta

35.partition

In [40]:
```
a="The art of begin alone"
print(a.partition("begin"))
print(a.partition("apple"))
```
('The art of ', 'begin', ' alone')
('The art of begin alone', '', '')

Practice Question

1.Write a Python program to reverse a string.

In [44]:
```
a="Python"
print(a[::-1])
```
nohtyP

2.Write a Python program to check whether a string is a palindrome.

In [47]:
```
a="madam"
b=a[::-1]
print(a==b)
```
True

3.Write a Python program to count vowels a string.

In [50]:
```
a="reeta"
res = a.count("a")+a.count("e")+a.count("i")+a.count("o")+a.count("u")
print(res)
```
3

4.Write a Python program to count the frequency of each character in a string.

In [54]:
```
a="aabacdefa"
b=a.count("a")
print(b)
```
4

5.Write a Python program to convert a string into uppercase..

In [58]:
```
a="abcdefghijklmnopqrstuvwxyz"
b=a.upper()
print(b)
```
ABCDEFGHIJKLMNOPQRSTUVWXYZ

6.Write a Python program to convert a string into lowercase.

In [61]:
```
a="ABCDEFGHIJKLMNOPQRSTUVWXYZ"
b=a.lower()
print(b)
```

abcdefghijklmnopqrstuvwxyz

7.Write a Python program to count the number of words in a string.

In [64]:
```
a="The art of begin alone"
b=a.split()
print(len(b))
```
5

8.Write a Python program to remove all spaces from a string.

In [69]:
```
a="  Reeta"
b=a.strip()
print(b)
```
Reeta

9.Write a Python program to replace spaces with hyphens in a string.

In [74]:
```
a="Reeta Manoj"
b=a.replace(" ","-")
print(b)
```
Reeta-Manoj

10.Write a Python program to check whether a string starts and ends with the same character using string slicing.

In [77]:
```
a="Python"
b=a.startswith("P")
c=a.endswith("n")
print(b)
print(c)
```
True
True